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Experimental study of saturated pool boiling heat transfer with FeCrAl- and Cr-layered vertical tubes under atmospheric pressure

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ABSTRACT

This study compares the nucleate boiling heat transfer coefficient (NBHTC) and critical heat flux (CHF) of two candidate coatings, FeCrAl and Cr, being considered for accident-tolerant fuel (ATF) cladding applications. To form an intrinsic surface roughness, the tube surfaces were initially grinded with 800 and 60 grit sandpapers, and then, FeCrAl and Cr were physically deposited using the direct current magnetron sputtering technique. The FeCrAl- and Cr-layered surfaces became hydrophilic on the lumped nanostructures and superhydrophilic on the particulate nanostructures, respectively. When subjected to the pool boiling conditions of vertically-oriented tubes, the NBHTC and CHF of the FeCrAl-layered tube increased by 24% and 34%, respectively, with the exception of the similar NBHTC generated by the 60 grit sandpaper. In contrast, the NBHTC of the Cr-layered tube decreased by 15% due to the suppressed nucleation on the particulate nanostructures, while CHF increased by 27%. The CHF enhancement was analyzed based on the liquid spreading behavior of a water droplet from the morphological changes. The capillary flow rate, which is a product of the liquid spreading rate, u_s, and arithmetic roughness height, R_a, resulted in a more accurate prediction of the CHF than the equilibrium contact angle. Additionally, the potential boiling performance of the FeCrAl and Cr coatings were discussed by comparing the pool boiling results of the vertical tube and horizontal plate orientations.

1. Introduction

Following the hydrogen risk generated by the existing nuclear fuel system of the Fukushima accident, great attention has been focused on the development of functional accident-tolerant fuel (ATF) claddings [1]. The first priority of the ATF requirements is to suppress the high temperature steam reaction, resulting in the reduction of hydrogen generation. Through abundant experimental testing regarding the improved oxidation resistance, several candidates are available to achieve this objective, such as stainless steel [2], FeCrAl [2–4], and Cr [5,6].

Surface coatings are a practically deployable means of applying these candidate materials to the nuclear fuel cladding system because maintaining the current cladding material (a zirconium-based alloy) is essential for preserving the neutron economy. In this regard, technical focus has been placed on testing the mechanical and thermal functions of the coating and improving the adhesion strength between the candidate material and body material in high temperature steam environments [5,6].

However, the surface coating encounters a boiling issue because the typical coating process inevitably retains surface structures at the micro/nanoscale that are capable of generating significant changes in the boiling performance. Over the past few decades, numerous researches have demonstrated that the surface structures apparent at the micro/nanoscale are able to delay a boiling crisis and improve boiling efficiency. The microscale structures possess several micro-cavities that increase the nucleation site density [7–11], and moreover, the micro/nanoscale structures enhance the liquid wicking characteristics, which effectively delay the formation of irreversible dry patches due to the additional momentum of rewetting liquid [12–14]. Thus, improving the surface structure design directly leads to achievable enhancement of the nucleate boiling heat transfer coefficient (NBHTC) and critical heat flux (CHF).

Although remarkable boiling enhancements (up to 200%) on the micro/nanostructured surfaces have been recently reported [15,16], the microscale roughness and porous structures may exhibit inappropriate properties for nuclear fuel cladding surface applications. This is because the microstructures and porous

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Nomenclature

V	voltage (V)
I	current (A)
D	diameter (mm)
L	length (mm)
r	radius (mm)
r_c	cavity mouth radius (μm)
P	pressure (Pa)
R_a	arithmetic roughness height (nm)
R_{sm}	arithmetic roughness pitch distance (μm)
q''	heat flux (kW/m^2)
K	surface factor
h_{fg}	latent heat (kJ/kg)
g	gravitational acceleration (m/s^2)
Q	wicked volume rate (m^3/s)
A_c	cross-sectional area of microflow channel (m^2)
A_{heated}	heated area (m^2)
u	velocity (m/s)
h_{pv}	peak-to-valley height (μm)
N_c	number of microflow channel
$D_{contact}$	length between the liquid droplet edges (mm)
$l_{contact}$	length of the contact diameter (mm)

Greek letters

ρ	density (kg/m^3)
σ_{lv}	surface tension of water (N/m)
θ	contact angle (°)
ϕ	heater orientation (°)

Subscripts

o	outer
e	equilibrium
r	receding
v	vapor
l	liquid
t	total
s	spreading
c	channel
w	wicking
$grav.$	gravitational
CHF	critical heat flux

medium are vulnerable to surface oxidation and easily lose their structural integrity in the thermal-hydraulic conditions of the core [17]. Thus, the candidate surfaces should essentially satisfy the following requirements for nuclear-level cladding surface design: the coating material is an ATF candidate, the surface roughness is lower than the technical limit (< 0.3 in R_a), and the surface structures and coated layer are not porous.

Few studies have investigated the surface effects of ATF cladding on the boiling performance while satisfying the aforementioned requirements. Seo et al. [18] introduced an FeCrAl coating fabricated by the direct current (DC) magnetron sputtering technique. Four types of FeCrAl structures were prepared by controlling the sputtering temperature to 25, 150, 300, and 600 °C. All of the FeCrAl coatings enhanced the CHF by approximately 14–42%, which indicates that the sputtering conditions are capable of changing the surface structures, thus, resulting in boiling enhancement. Also using the DC magnetron sputtering technique, Son et al. [19] fabricated Cr coatings on different grinded surfaces, thereby, investigating the roughness-augmented CHF enhancement. The CHF was enhanced by 79% with a surface roughness increase of 0.25 μm for R_a. Thus, the results emphasized the importance of the intrinsic surface roughness on the CHF enhancement and sputtering conditions. Kam et al. [20] examined the pool boiling CHF of an SiC-coating, Cr-coating, and bare Zircaloy-4. Compared to the bare Zircaloy-4, the CHF of the SiC coating increased by 150%, whereas the CHF of the Cr coating decreased by 70%. They concluded that the higher CHF with the SiC coating is a result of the improved wettability in comparison to the Cr coating. By applying the deposition of Cr_2O_3 nanoparticle, Cr_2O_3 coating (RF sputtering), and Cr coating (DC sputtering) to wire surfaces, Son et al. [21] showed that pool boiling CHF increases by 6%, 88%, and 93%, respectively. The major cause of CHF enhancement was originated from improved capillary action on modified surface structures. However, these studies focused on the pool boiling performance of horizontal plates and wire unlike the real fuel cladding geometry. Given that practical nuclear claddings are aligned with vertically-oriented tubes, a more realistic situation should be introduced, because the boiling performance significantly depends on the heater geometry and orientation [22,23]. Moreover, the candidate materials require a comparison at the same coating technique and boiling conditions.

This study compares the boiling heat transfer performance of FeCrAl- and Cr-layered vertical tubes. The surface coating was generated using the DC magnetron sputtering technique, which secures columnar film growth without generating porous structures. In addition, to investigate the roughness contribution to the boiling performance, the tube surface roughness was varied below 0.3 μm for R_a. The NBHTC and CHF of the FeCrAl- and Cr-layered tubes vertically heated were quantified under the pool boiling condition. Importantly, the CHF variation with the liquid spreading behavior was analyzed in order to understand the role of sputtered and roughened structures on the CHF enhancement. Finally, based on the results of previous research and those presented herein (*i.e.* previous horizontal boiling [18,19] and current vertical boiling, respectively), we discuss the potential boiling performance of the FeCrAl and Cr coatings for practical applications.

2. Experiments

2.1. Surface preparation

Prior to implementing the DC magnetron sputtering technique, the tube surfaces were initially grinded with 800 and 60 grit sandpaper in an attempt to examine the effects of intrinsic surface roughness on boiling enhancement. The grinded tube surfaces contain several micro-scratched lines, and their cross-sectional dimension has the potential to cause capillary wicking flow [19,24]. The FeCrAl and Cr depositions were then performed using the DC magnetron sputtering technique. The sputtering conditions are summarized in Table 1. In our previous study [18], it was found that the CHF of FeCrAl-layered surfaces increased non-monotonically with the substrate temperature of 25, 150, 300, and 600 °C. The highest CHF was observed at the substrate temperature of 150 °C because its roughness factor was the highest among the simulated substrate temperatures. Moreover, the roughness factor decreased as the exposure time increased from 1 h to 6 h. In this regard, the sputtering process was conducted at a substrate temperature of 150 °C for 1 h.

Table 1
DC sputtering conditions.

Sputtering parameter	Target condition
Target material	FeCrAl (75:20:5 wt%) Cr (99.95% purity)
Substrate material	Stainless steel grade 316L tube
Substrate temperature (°C)	150
Exposure time (hour)	1
Base pressure (Torr)	1×10^{-5}
Working pressure (Torr)	1×10^{-2}
DC power (W)	150–160
Inert gas flow	Ar at 29.7 sccm

Fig. 1 shows an illustration showing the overall DC magnetron sputtering adopted in this study. The DC magnetron sputtering system includes a target material assembled with magnets and a DC gun, a substrate supporting plate, halogen heaters, and an argon gas (Ar) injection system. The Ar+ particles, which are initially ionized by a high potential difference, are used to bombard the target material. The energetic sputtered atoms deposit onto the slowly-rotating tube surface (~3 rpm). During the deposition of the sputtered atoms, the adatoms (the atoms physically deposited on the substrate) are densely distributed along the micro-scratched lines and form different nanoscale clusters, for which the morphology changes based on the sputtering parameters.

Fig. 2(a) and (b) show the coated layer thickness and change of the surface characteristics, respectively, before and after the DC sputtering process. After a film growth of approximately 1.5 μm in t hickness, the surface wettability changed according to the target material. The equilibrium contact angle of the FeCrAl-layered surface decreased by approximately 29–35° due to the presence of the lumped nanostructure. The particulate nanostructures (~200 nm in size) on the Cr-layered surfaces caused complete wetting (~0°), thus, signifying that the particulate nanostructures tend to cause further wetting of the water droplet. A comparison of the extended contact line of the nanoscale Cr particulates and lumped-FeCrAl structures indicates that the nanoscale Cr particulates provide increased surface tension force [25] and/or nanoscale capillarity [19,26].

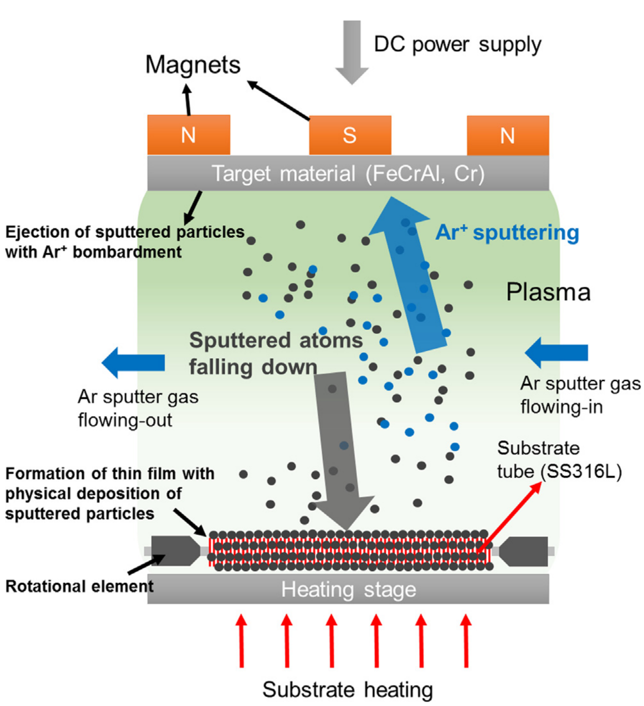

Fig. 1. A schematic of the DC magnetron sputtering process.

The change of the surface roughness is shown in Fig. 3(a). Although the surface roughness increases for all of the cases with lower sandpaper grit numbers, the roughness increase on the Cr-layered surfaces is not remarkable when compared to the bare and FeCrAl cases. This is because for the Cr sputtering process adatoms agglomerate together to form particulate clusters up to 200 nm in size (see the scanning electron microscope image of Cr-layered surface in Fig. 2(b)), and then, the micro-scratched lines, which are a few micrometers in size for the 60 grit case, can be filled with smaller Cr particulates.

The aforementioned results indicate that the surface wettability and roughness vary based on the selected target material, despite the same sputtering conditions. This is possible because the growth mechanism of the metal films for the DC sputtering process is determined by the structure zone model [27], which largely depends on the deposition parameters, such as the inert gas pressure and substrate temperature ratio [27]. Importantly, the substrate temperature ratio is defined as the substrate temperature divided by the melting point of the target material, and thus, the different melting temperatures of FeCrAl and Cr can generate morphological changes during the DC sputtering process. Considering that the typical melting point of pure Cr (~1900 °C) is higher than the series of FeCrAl alloys (~1400 °C), the surface structure of the Cr layer becomes increasingly particulate at the lower substrate temperature ratio of the sputtering conditions used herein. This morphological difference was also found in our previous sputtering research [18,19]. In this regard, it should be noted that the material and morphological effects are not completely separated in this study because the micro/nanostructure on coated surfaces is changed with a type of target material. Instead, this study investigates what kind of material will be more beneficial for generating micro/nanostructures, which are favorable to boiling enhancement even at the same coating technique and its working conditions.

Although significant changes in the surface wettability and roughness of the FeCrAl and Cr coatings were obtained, the parametric relation between them was very weak, as shown in Fig. 3 (b). Firstly, a small decrease in the contact angles of 7° and 6° were observed in the bare and FeCrAl cases, despite significant increases of 8 and 4 times in roughness, respectively. Moreover, the contact angle for the Cr case was indiscernible with the change in surface roughness because all of the surfaces after the Cr coating became superhydrophilic, regardless of the surface roughness. Since this result causes difficulty in quantitatively characterizing the contribution of roughness on the contact angle variation, another approach is required to analyze the change in the surface wetting characteristics with varying roughness.

2.2. Liquid spreading experiment

The contact angle, which quantitatively represents the surface wettability, has been determined as an effective indicator for analyzing the boiling heat transfer, especially regarding CHF enhancement [28,29]. However, the contact angle has a limitation in distinguishing the wicking capability for different surface structures. In previous research [19,30], it was confirmed that the equilibrium contact angle is equally 0° for different surface roughness values, whereas prior to reaching the equilibrium state, the spreading rate of the water droplets increases with surface roughness. Likewise, in this study, the equilibrium contact angles of the FeCrAl- and Cr-layered surfaces were hardly changed as the surface roughness increases. Thus, to clearly distinguish the effects of the surface roughness on the wetting and wicking behaviors, a liquid spreading experiment was introduced herein.

The liquid spreading behavior was observed using the EasyDrop goniometer (made by Krüss) coupled with a high-speed camera (Phantom v7.3) having a frame rate of 1000 frames per second.

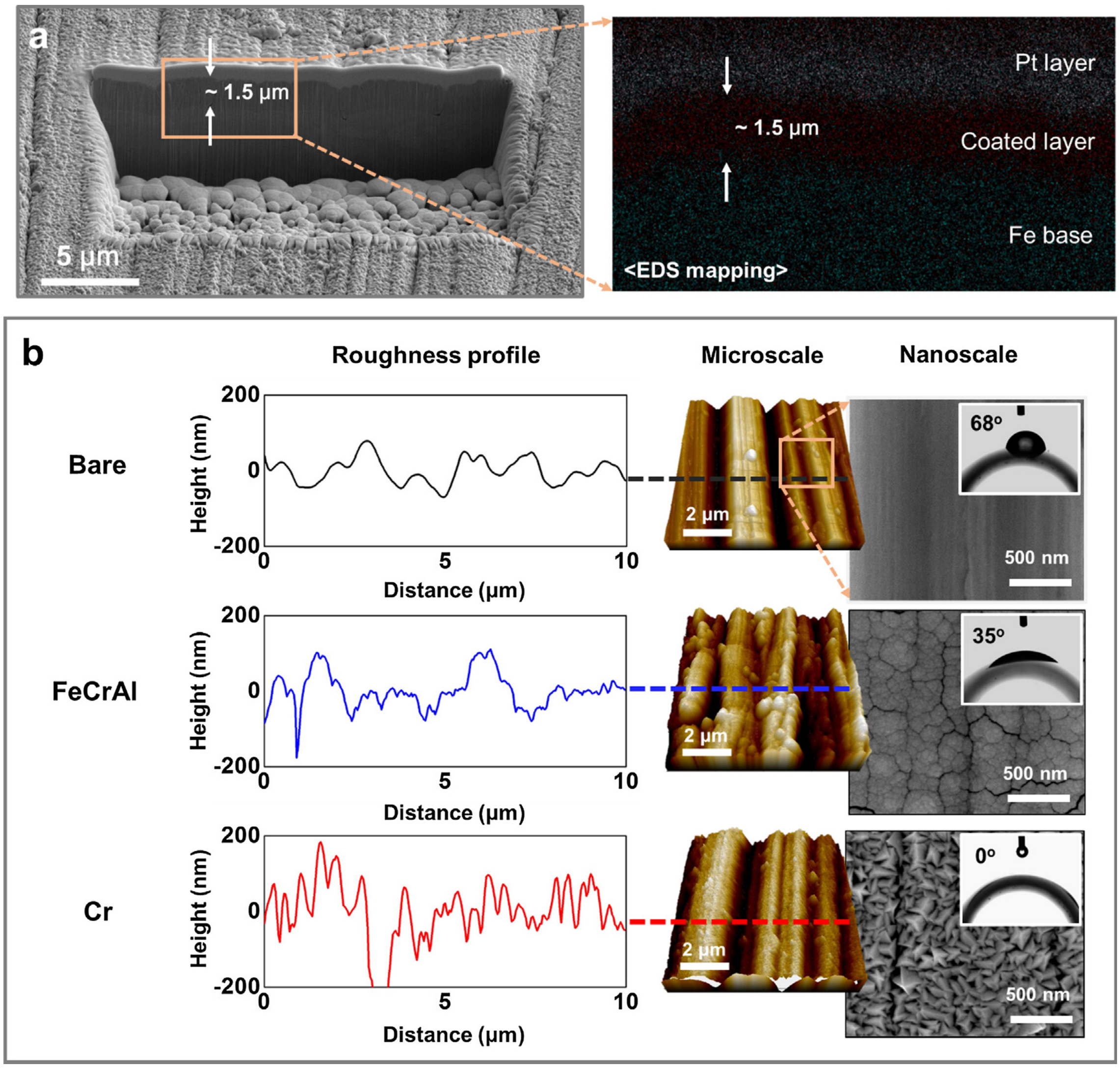

Fig. 2. (a) Identification of coated layer thickness using focused ion beam (FIB) and energy dispersive spectroscopy (EDS) mapping analysis. Herein, a Pt layer was used only for utilizing the FIB system. (b) Different surface characteristics, including the equilibrium contact angle and surface morphologies at the micro- and nano-scales. In particular, roughness profiles shows an increase of surface area ratio in order of Cr-layered, FeCrAl-layered, and bare surfaces.

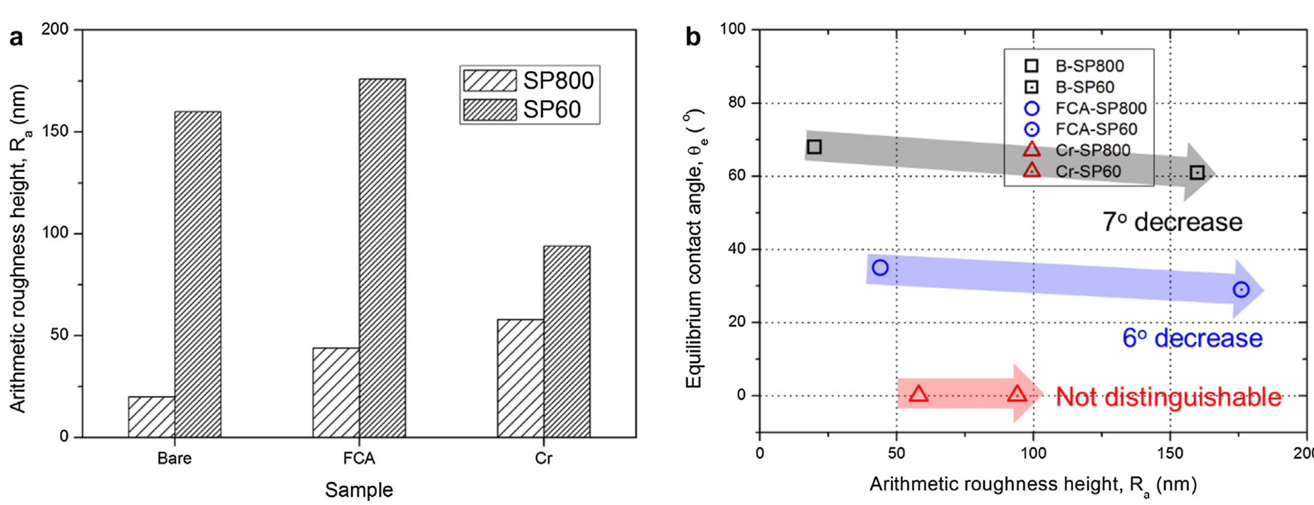

Fig. 3. (a) Change of the surface roughness and (b) weak relation between the surface roughness and equilibrium contact angle. In this paper, grit number of sand paper is abbreviated to SP. (Note that FeCrAl has been abbreviated as FCA in all figures.)

Fig. 4 shows measurement system used for quantifying the liquid spreading trends. To observe the liquid spreading behavior along the micro-scratched lines, the visualization direction was established as normal to the direction of scratching, as shown in Fig. 4(b). Thus, the movement of the contact line can be approximated as the amount of liquid spreading along the micro-scratched lines. Fig. 4(c) shows a sample of the water droplet spreading trends on the bare surface. To obtain the actual contact length between the water droplet and tube surface, the arc contact length, $l_{contact}$, is calculated as follows

$$l_{contact} = D_0 \arcsin\left(\frac{D_{contact}}{D_0}\right) \tag{1}$$

where D_o is the tube outer diameter, and $D_{contact}$ is the measured diameter. Using this method, the liquid spreading behavior is compared for the bare, FeCrAl-, and Cr-layered tube surfaces in Section 3.2.

2.3. Pool boiling experiment

Fig. 5(a) and (b) show the pool boiling apparatus and detailed design of the test section. The Joule heating method was adopted to generate the resistive heat along the tube. Two K-type thermocouples (TCs) were inserted into the TC guide tube and maintained at a distance of 10 mm from the top and bottom. To quantify the NBHTC and detect the CHF, the top-located TC was used, because the CHF repeatedly occurred at this location. The empty space between the tube heater and TC guide tube was filled with a high temperature epoxy (a mixture of SamplKwick™ 20-3566 and 20-3568) to provide thermal and electrical insulation. The tube length, outer diameter, and tube thickness are 100, 9.52, and 0.88 mm, respectively. The active heated length is 80 mm due to the lengths of 10 mm at each end that were used for the electrical contact between the tube heater and copper electrodes.

Prior to applying the steady-state heat flux to the tube heater, the deionized water was saturated at atmospheric pressure using the preheaters immersed in an isothermal bath. To avoid the degassing process during measurement, the saturation state was maintained for more than an hour, during which the non-condensable gases that would impact the bubble dynamics were removed. When bubbles were not observed in the isothermal bath, the steady-state heat flux was carefully applied in steps of 50–60 kW/m^2 until the heat flux reached 70% of the predicted CHF, as shown in Fig. 5(c). After this point, a smaller amount of heat flux (approximately 20–30 kW/m^2) was applied to prevent sudden formation of vapor film. During each heat flux increment, the steady-state condition was maintained for a minimum of two minutes, while the steady wall temperature readings were confirmed. Finally, a sudden increase of the inner wall temperature and sequential formation of the vapor film were simultaneously observed when the applied heat flux reached the CHF, as shown in Fig. 5(c) and (d), respectively. As the vapor film propagates along the heated tube, an excessive heat load occurs, which generates a fatal impact on the pool boiling system, and the DC power supply was deactivated within 3–5 s.

For each time step, the voltage and current signals were collected by a cDAQ device (National Instruments) in order to calculate the applied heat flux, as shown in Eq. (2).

$$q'' = \frac{Power}{A_{heated}} = \frac{\Delta VI}{\pi D_o L_{heated}} \tag{2}$$

where ΔV is the voltage drop across the length of the heated tube, I is the applied current, D_o is the outer diameter of the tube, and L_{heated} is the heated length. Given the independent variables, V, I,

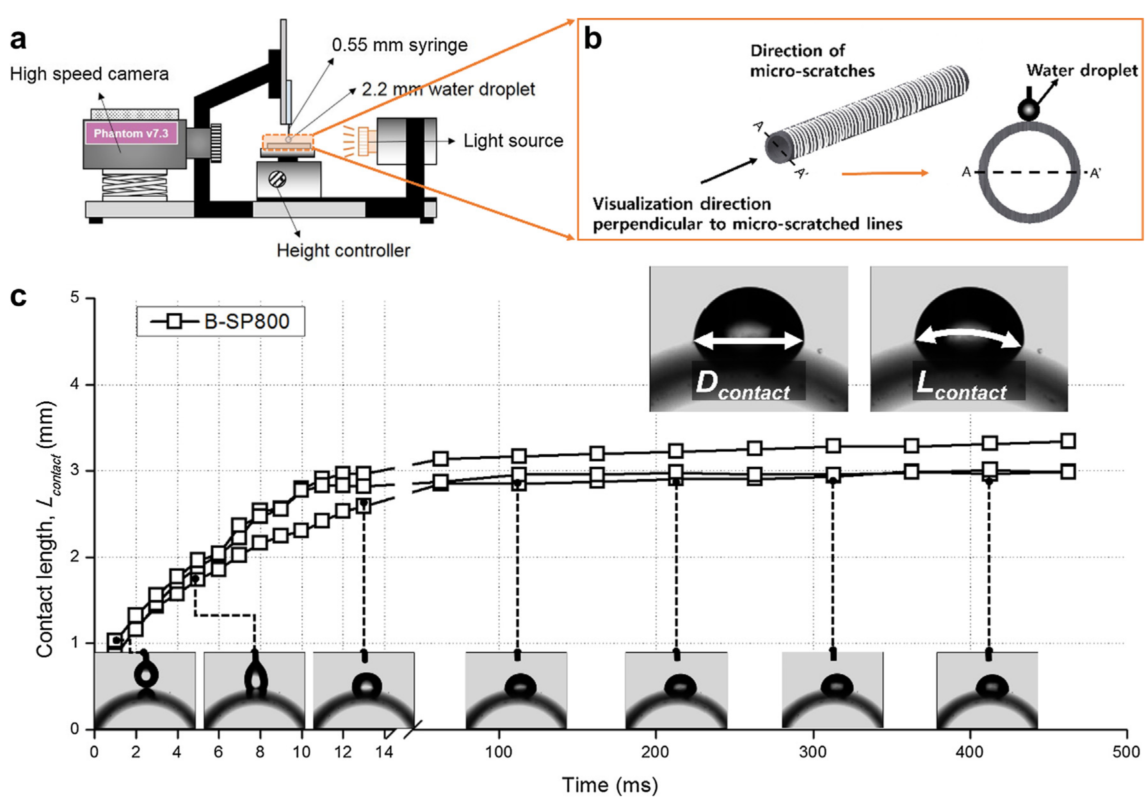

Fig. 4. Measurement of the liquid spreading capabilities: (a) goniometer (EasyDrop made by Krüss) coupled with a high-speed camera (Phantom v7.3), (b) alignment direction, and (c) sample data trend of the increasing contact length, $L_{contact}$, on the bare surface.

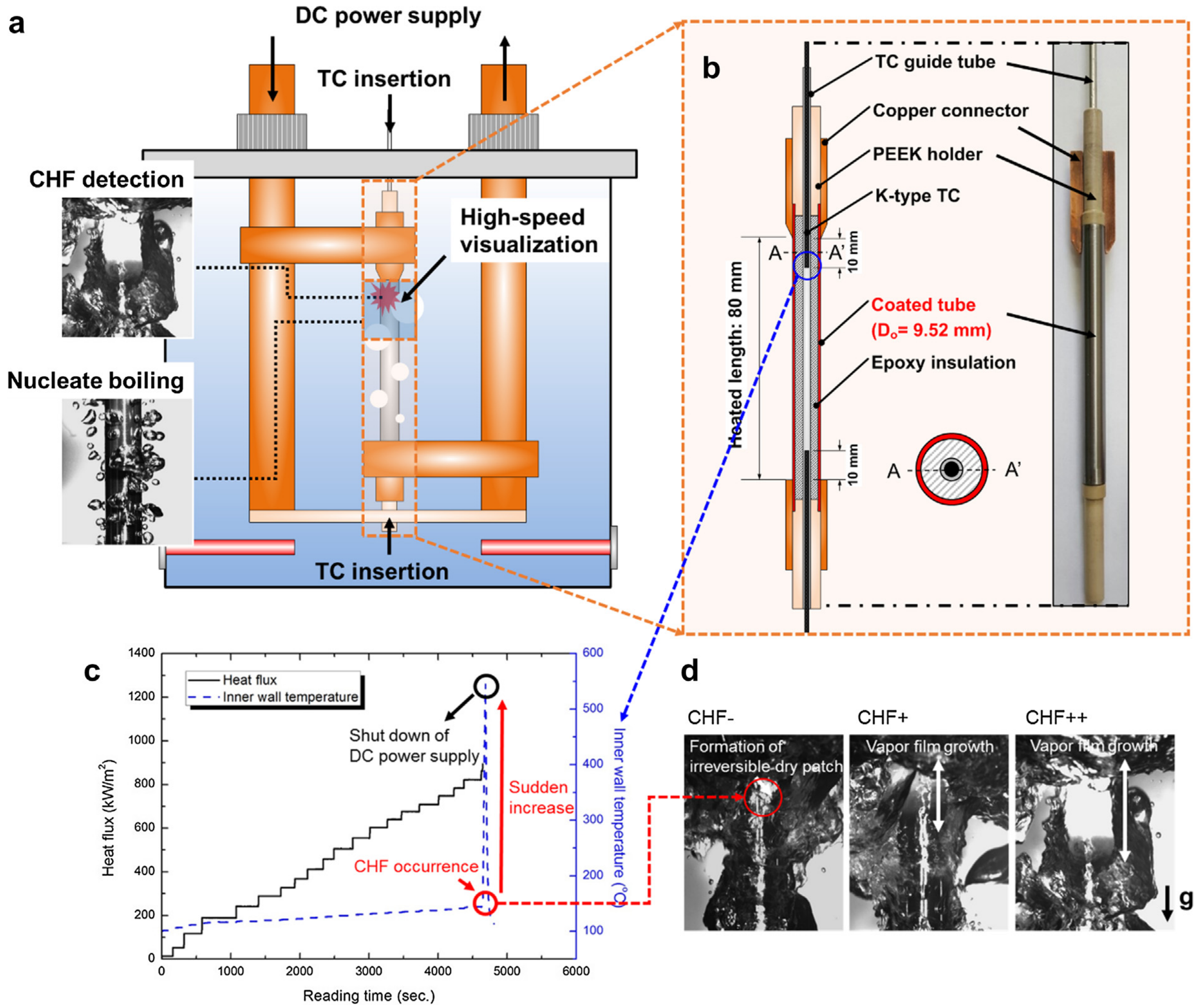

Fig. 5. (a) Schematic of the pool boiling apparatus, (b) schematic of the test section, (c) time sequential profiles of the applied heat flux and inner wall temperature, and (d) high speed images of the vapor film propagation at the CHF.

D_o, and L_{heated}, the heat flux measurement uncertainty was calculated using an error propagation method, as given by Eq. (3).

$$\left(\frac{U_{q''}}{q''}\right)^2 = \left(\frac{U_V}{V}\right)^2 + \left(\frac{U_I}{I}\right)^2 + \left(\frac{U_D}{D}\right)^2 + \left(\frac{U_L}{L}\right)^2 \quad (3)$$

The uncertainties of the voltage (U_V/V), current (U_I/I), width (U_D/D), and heated length (U_L/L) were less than 0.5, 0.5, 2.2, and 5.0%, respectively, and thus, the maximum heat flux measurement uncertainty, $U_{q''}/q''$, for the heat flux was calculated to be 5.5%. Since DC Joule heating was used to secure a uniform heat flux profile, this heat flux uncertainty is equally applied along the heated length. However, the temperature profile is not uniform along the heated length due to the variation of vapor quality, which generates a variable NBHTC along the tube axial direction. As a means of comparing the NBHTC at the same location, the boiling curves are reported based on the inner wall temperature, which is measured at the location of the blue circle shown in Fig. 5(b).

3. Results and discussion

3.1. Comparison of the nucleate boiling heat transfer between the FeCrAl- and Cr-layered surfaces

In Fig. 6, the NBHTC of FeCrAl-SP800 increased by 24% at the high heat flux (~636 kW/m²), whereas the NBHTC of Cr-SP800 decreased by 10%. In the case of SP60, the effect of the FeCrAl layer on the NBHTC was indiscernible, which is a similar trend to that of SP60. Contrary to FeCrAl-SP60, the NBHTC of Cr-SP60 was approximately 15% lower than that of SP60. Since only the coating material and structure were altered, the difference of the NBHTC can be explained by the surface wettability and roughness, which were changed before and after the surface grinding and coating.

With respect to bubble nucleation, the surface wettability and roughness correlate with the Young-Laplace equation, which describes the pressure difference, P_v-P_l, between the vapor bubble, P_v, and surrounding liquid, P_l, as shown in Eq. (4):

$$P_v - P_l = \frac{2\sigma_{lv}\cos\theta_e}{r_c} \quad (4)$$

where σ_{lv} is the liquid surface tension, θ_e is the equilibrium contact angle, and r_c is the cavity mouth radius. In this equation, both the decrease of the equilibrium contact angle and size of cavity mouth result in an increase of the nucleation pressure, thus, requiring additional wall superheat (*i.e.* increased heat flux) to actively nucleate the superheated liquid inside the cavities. In a single bubble nucleation experiment, S. Witharana et al. [31] demonstrated that the Young–Laplace equation agreed with their experimental data, which varied with the size of the cavity mouth.

Fig. 7(a) shows the calculated Young-Laplace pressure differences for the coating materials and according to the grit number of sandpaper. Firstly, a higher surface roughness results in a lower

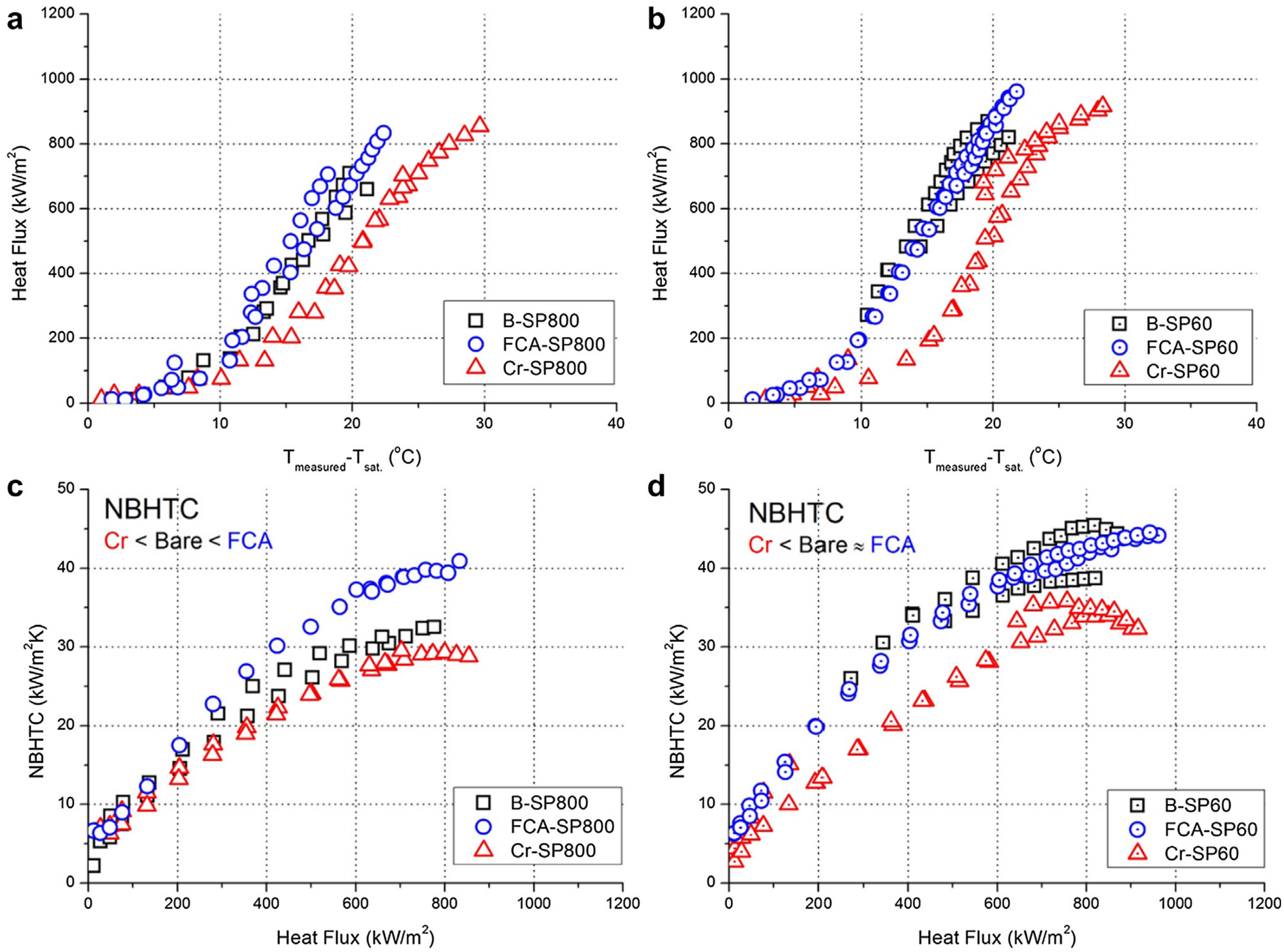

Fig. 6. Pool boiling curves for (a) SP800 and (b) SP60. NBHTC trends for (c) SP800 and (d) SP60.

pressure difference because the size of cavity mouth statistically increases with an increase in surface roughness. This result is consistent with M.G. Kang's study [7], in which a higher surface roughness generates additional cavities, thus, increasing the nucleate site density. In the comparison of the coating materials, the pressure difference shows an increasing trend for the bare, FeCrAl, and Cr layers, respectively, which indicates that increasing the surface wettability suppresses bubble nucleation. This negative

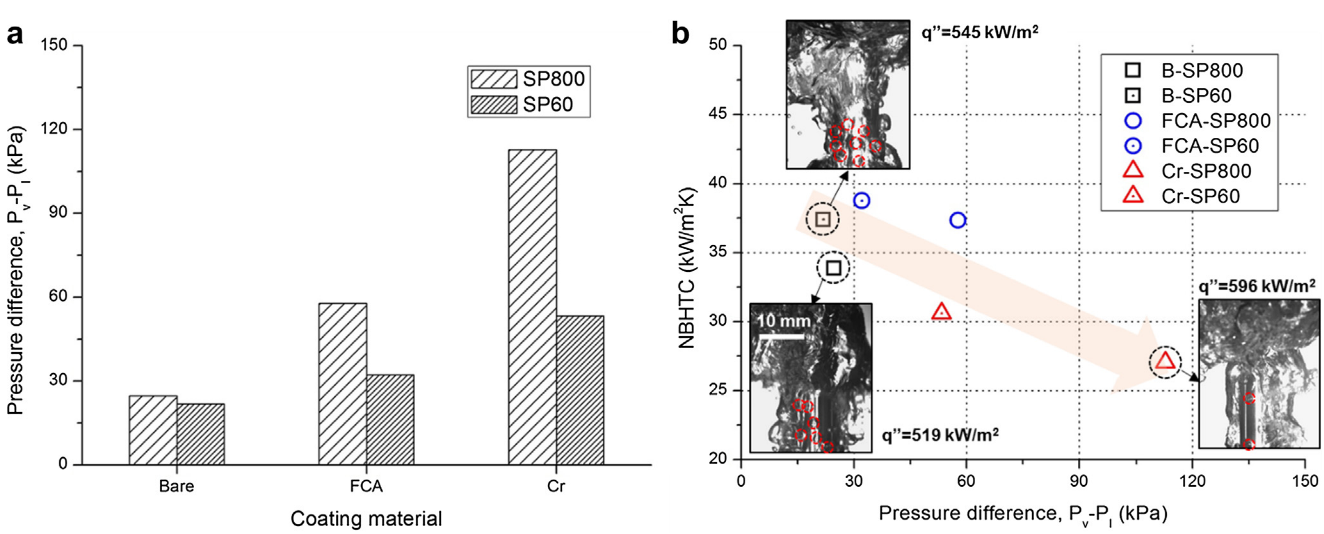

Fig. 7. Comparison of the nucleate boiling performance: (a) Young–Laplace pressure difference between the coating materials and (b) decreasing trend of the NBHTC with the Young–Laplace pressure difference. For selected cases of interest, inset of high-speed images exhibits the difference between numbers of active nucleation, which is marked as a red-dashed circle.

relation has been previously described by Wang and Dhir [11], in which the active nucleate site, n'', is proportional to 1-cosθ_e. Fig. 7(b) shows that the NBHTC clearly decreases with an increasing Young-Laplace pressure difference. Moreover, visual observations exhibited as the inset in Fig. 7(b) support the decreasing NBHTC trend based on generating fewer number of active nucleation in Cr-SP800 than in bare cases. These results demonstrate that the superhydrophilic nanostructure of the Cr-layered surface is unfavorable for bubble nucleation when compared to the hydrophilic surfaces of the bare surface and FeCrAl coating. Kim et al. [30] reported similar results in their research, where they compared the NBHTC with R_a of a superhydrophilic micro/nanostructure (aluminum-coated) and hydrophilic microstructure. The results showed that the NBHTC of the superhydrophilic micro/-nanostructure was consistently lower than that of the hydrophilic microstructure for a given surface roughness. In this manner, superhydrophilicity with nanoscale roughness is disadvantageous to the nucleate boiling efficiency.

Regardless of the NBHTC reduction, when compared to the CHF of B-SP800, the CHFs of the FeCrAl- and Cr-layered surfaces were enhanced by approximately 34% and 27%, respectively. Moreover, increasing the surface roughness caused additional CHF enhancement. However, given this CHF enhancement, there was no discernible benefit of the superhydrophilic Cr coating when compared to the hydrophilic FeCrAl coating, as both CHFs were similar.

In Fig. 8, the present data are compared with the Kandlikar model [28] and open source data [18,19]. The Kandlikar model mainly considers the single bubble force balance involving the vapor recoil momentum and liquid surface tension at the apparent triple contact lines. Eq. (5) of the Kandlikar model indicates that the CHF increases with a decrease in the contact angle.

$$q''_{Kandlikar} = K \times h_{fg}\rho_v^{1/2}[\sigma_{lv}g(\rho_l - \rho_v)]^{1/4}$$
$$K = \left(\frac{1+\cos\theta_r}{16}\right)\left[\frac{2}{\pi} + \frac{\pi}{4}(1+\cos\theta_r)\cos\phi\right]^{1/2} \quad (5)$$

where h_{fg} is the latent heat, ρ_l is the liquid density, ρ_v is the vapor density, and g is the gravitational acceleration. The surface factor, K, consists of the receding contact angle, θ_r, and heater orientation, ϕ. The trend of the results also indicates that an increasing CHF corresponds to a decrease in the contact angle; however, the enhancement ratio was lower than that of the plate-type heaters due to the orientation effect, ϕ, included in Kandlikar model. The CHF data were similar to the Kandlikar model with respect to the surface wettability and heater orientation. However, the equilibrium contact angle does not solely clarify the cause of the CHF enhancement, because the Kandlikar model only considers the surface wettability without the roughness parameters. In order to clearly determine the effects of both parameters on the CHF enhancement, we will analyze the causes of CHF enhancement in the following section, focusing on the liquid spreading behavior with morphological changes.

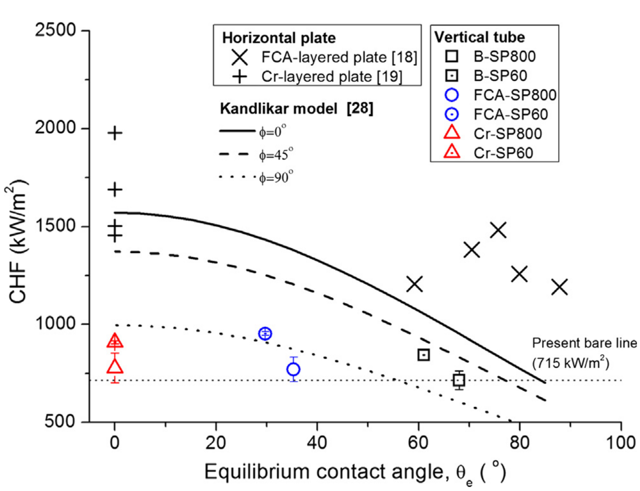

Fig. 8. CHF comparison with the Kandlikar model [28] and open source data (FeCrAl layer [18] and Cr layer [19]). Note that the sputtering conditions of the open source data (plate sputtering) are different than the present conditions (tube sputtering), although the DC magnetron sputtering technique was applied.

3.2. Effect of the liquid spreading on CHF enhancement

Recently, direct observations of the boiling structure using an infrared camera [12] and synchrotron X-ray [14] have emphasized the importance of capillary wicking on CHF enhancement, where the cooling liquid around the dry spots is sucked into the surface structures. This liquid momentum can be passively activated by the capillary wicking inside micro/nanostructures. In this regard, considering that the effects of surface wettability and roughness have been presented herein, the CHF enhancement also requires analysis that considers the influence of capillary wicking.

Fig. 9 shows a simple schematic of the evaporative liquid distribution around a dry spot. The growth of the dry spot primarily competes with the liquid inflow. Thus, if additional liquid inflow is provided, a greater amount of cooling liquid must be evaporated to cause local dryout, which results in CHF enhancement. This additional enhancement is generated by the wicking heat flux, $q''_{wicking}$, and its relation to the capillary wicking flow inside solid structures is expressed in Eq. (6).

$$q''_{wicking} = \frac{\rho_l Q_t h_{fg}}{A_t} \quad (6)$$

where ρ_l is the liquid density, Q_t is the total volumetric flow rate, h_{fg} is the latent heat, and A_t is the bubble base area. In Eq. (6), $q''_{wicking}$ is proportional to Q_t for the given constants. Accordingly, quantifying the wicked volume rate is a key issue for evaluating the CHF enhancement.

In Fig. 8, Kim et al. [32] classified Q_t into the gravitational and wicking volumetric flow rates, $Q_{grav.}$ and $Q_{wicking}$, respectively. Here, $Q_{grav.}$ was expressed as the hydrodynamic head of a growing bubble, macrolayer size, and Rayleigh-Taylor instability, and, therefore, $Q_{grav.}$ was treated as a constant for all of the surfaces. This indicates that the volumetric liquid flow only varies with wicking behavior. Rahman et al. [16] and Ahn et al. [33] found that the wicked volume rate on textured surfaces is clearly proportional to the CHF enhancement ratio. Based on their results, in order to directly correlate the capillary wicking parameter with CHF enhancement, a combination of the capillary flow velocity and cross-sectional area of the surface structures requires consideration.

In this manner, a simple model of the microflow channels was generated using atomic force microscopy images in order to simulate the aforementioned wicking concept that is applicable to our

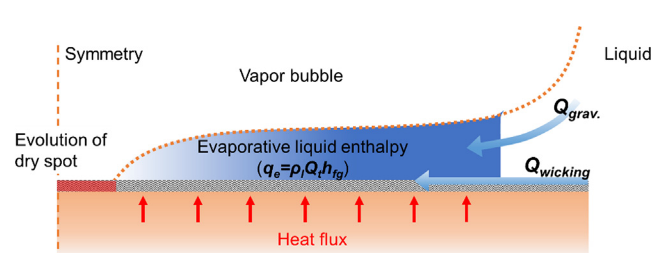

Fig. 9. Schematic of the liquid wicking contribution to dry spot rewetting.

surface design (randomly-roughened surfaces). In Fig. 10(a), the wicked volume rate, Q_t, can be expressed as Eq. (7).

$$Q_t = Q_{grav.} + Q_{wicking} \sim u_s A_c N_c \tag{7}$$

Here, u_s is the spreading rate of a water droplet, A_c is the cross-sectional area of the microflow channel, and N_c is the number of microflow channels. The right hand side of Eq. (7), $u_s A_c N_c$, represents the total amount of liquid that is sucked along triple contact line. This relation is apparent when the microscopic view of the hydrophilic surface structures is completely filled with the wicked liquid, which displays a type of liquid film flow [34]. However, if the peak-to-peak distance of the roughness widens at the micro-scale (*i.e.* lower structural density), this assumption becomes not acceptable, because the liquid meniscus gradually forms a concave shape due to both of the hydrophilic nature of solid structures and the vapor recoil force [14].

Next, assuming that the triangular morphology of the microflow channel continues along the micro-scratched lines, the cross-sectional area of the microflow channel, A_c, becomes proportional to a product of peak-to-valley height, h_{pv}, and peak-to-peak distance, both of which are expressed in Fig. 10(b). Based on typical definition of roughness parameters, the peak-to-peak distance of the roughness profile corresponds to the mean width of the profile elements, R_{sm}, whereas R_a is always lower than h_{pv} because R_a is calculated by integrating all profile elements over a given sampling length. By comparing the measured h_{pv} with R_a values in Fig. 10(c), it was found that h_{pv} is approximately 3 times higher than R_a. In this regard, h_{pv} can be substituted for $3R_a$, which forms the relation of $A_c = 3R_aR_{sm}/2$.

Morphologically the number of a given microflow channel, N_c, corresponds to the number of peaks along the roughness profile. Thus, the number of the microflow channel, N_c, should be inversely proportional to R_{sm} in unit length. Since the spreading rate, u_s, distributes uniformly along the microflow channels, the relation between the volumetric flow rate, spreading rate, and structural parameters is expressed in Eq. (8).

$$Q_t \sim u_s A_c N_c = u_s \left(\frac{3}{2} R_a R_{sm}\right) \frac{1}{R_{sm}} \sim u_s R_a \tag{8}$$

To quantify the spreading rate, u_s, the increasing trend of the arc contact length with time was first generated, as shown in Fig. 11. Here, the spreading mode is clearly distinguished before and after the range of approximately 10–15 ms. Prior to the 10–15 ms range, a water droplet spreads rapidly due to the impact-driven inertia, which is governed by gravitational potential energy, kinetic energy, and interfacial surface tension energy [26,35]. However, following 10–15 ms range, in which aforementioned forces become weak due to a remarkable decrease in curvature of the droplet, the surface capillarity mainly drives the secondary spreading. This implies that the effect of surface roughness starts to be effective. This difference of the droplet spreading behavior between roughened surfaces was also observed in Kim et al.'s study [30], in which the surface roughness becomes effective to the secondary spreading rather than the primary spreading. Moreover, Wemp and Carey [26] explained that the primary spreading droplet footprint behaves synchronously with the wicking precursor line, while the secondary spreading capillary force causes hemi-spreading, where the wicking precursor line precedes the droplet

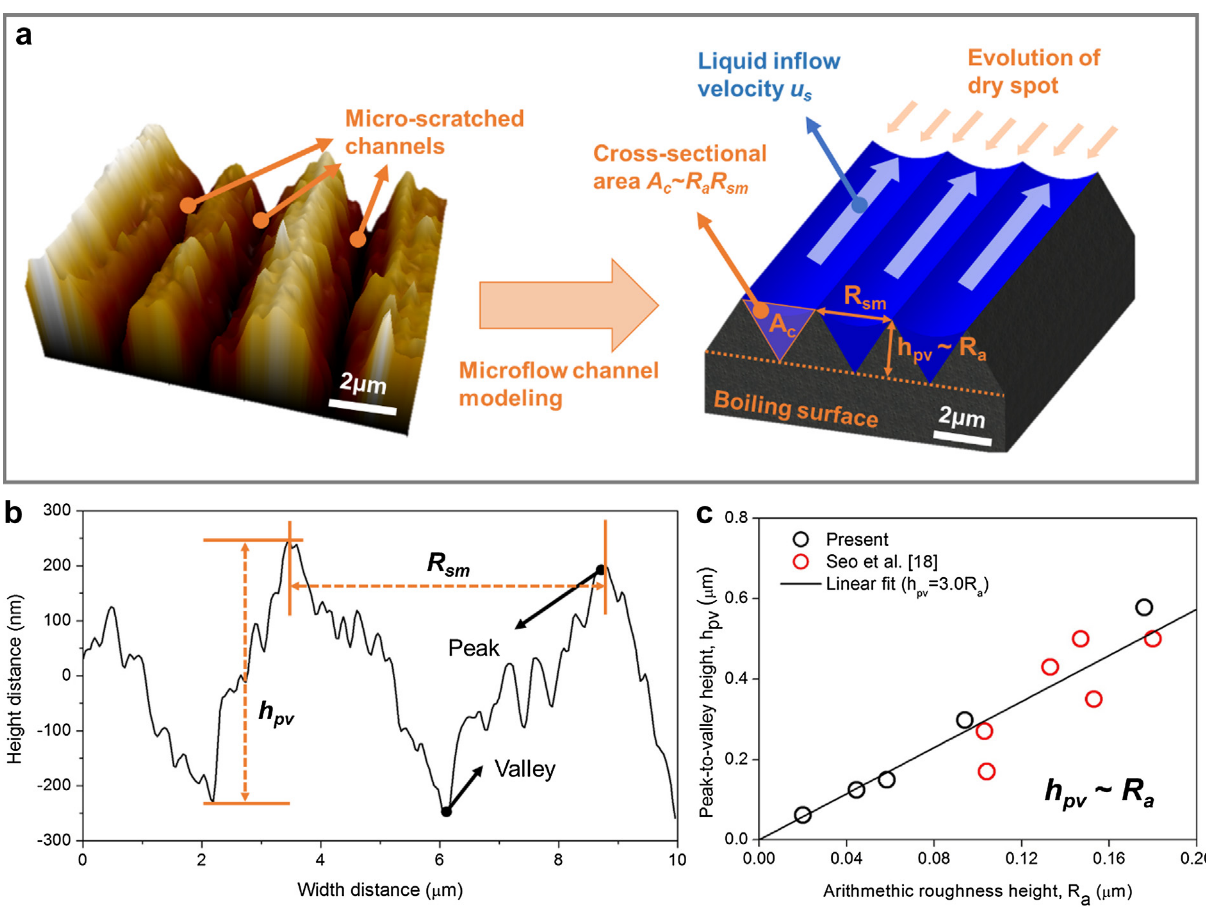

Fig. 10. (a) Microflow channel modeling for the randomly-roughened surfaces, (b) the identification of peak-to-valley distance, h_{pv}, along the representative roughness profile, (c) the linear relation between h_{pv} and R_a.

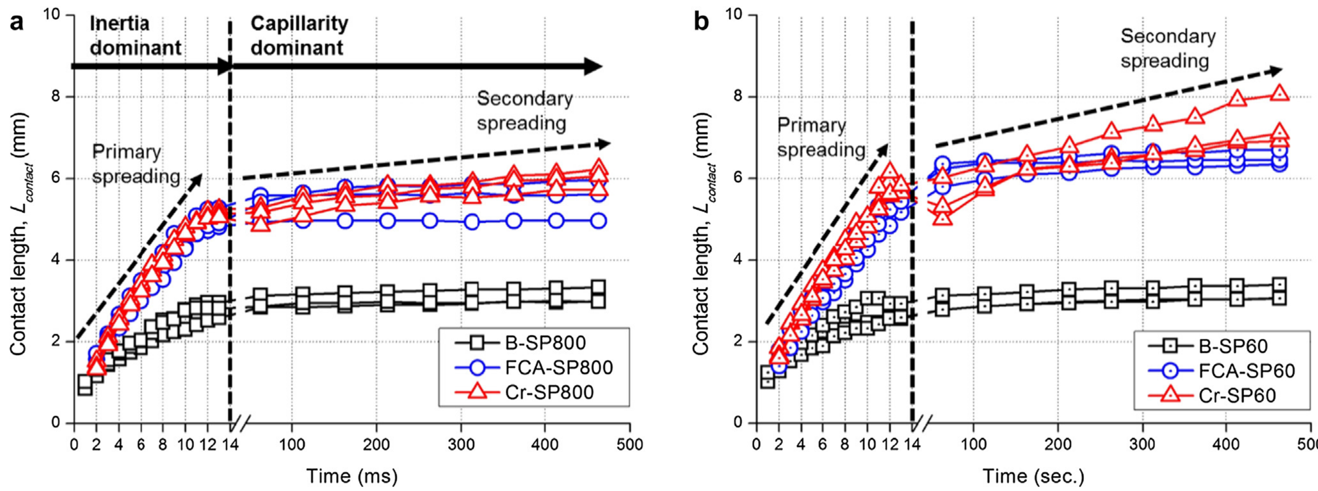

Fig. 11. Liquid spreading behavior. Increasing trends of the contact diameter, $L_{contact}$, with time for (a) the SP800 cases and (b) SP60 cases.

footprint. In our experiment, however, only synchronous spreading was observed for the primary and secondary spreading. This is because the capillarity of our surfaces seems to become less significant when hemi-wicking occurs. In this manner, we assumed that the synchronous spreading still dominates the secondary spreading, and the wicking front moves simultaneously with the droplet footprint.

The spreading rate was quantified by calculating the slope of the increasing arc contact length. In Fig. 12(a), totally, the spreading rate becomes increasingly faster for the Cr-layered, FeCrAl-layered, and bare surfaces, respectively, although the difference of the spreading rate between FeCrAl- and Cr-layered surfaces was weak for the primary spreading. Here, the significant increase of the spreading rate on the FeCrAl- and Cr-layered surfaces compared to that of bare surface could be due to an increase in interface surface tension energy resulting from the augmentation of contact line on sputtered-nanostructures (see the variation of roughness profiles in Fig. 2(b)). As previously mentioned, however, the effect of surface roughness on the droplet spreading is barely observed in this stage because the interface surface tension energy is still effective to decrease the curvature of the droplet [26,35]. Thereafter, the capillary force caused by the surface roughness intensifies the secondary spreading more than the primary spreading, considering that the relative difference of the spreading rate was higher in the secondary spreading (40–120%) than in the primary spreading (<20%), as shown in Fig. 12(b). In particular, the particulate-nanostructure on the Cr layer (~120% enhancement) results in a higher spreading rate than the lumped-nanostructure on the FeCrAl layer (~90% enhancement). In this regard, we are able to successfully distinguish the liquid wicking momentum with the surface roughness, the trends of which were not observable in the measurement of equilibrium contact angle. This result also supports that the liquid wicking behavior can differ with morphological changes, despite the same contact angle [19].

Combining the secondary spreading rate, u_s, with the arithmetic roughness height, R_a, in Fig. 13(a), the trend of the capillary flow rate from Eq. (8) correlates with the CHF data, which had an unpredictable trend when only considering the contact angle of Eq. (5). Fig. 13(b) shows that the parametric correlation between the CHF and capillary flow rate is very high, having an R-squared value of 0.95. This result is consistent with other relevant studies [19,30,36], which have also confirmed the current findings on the horizontal plate configuration. In addition to the previous studies, our approach further validates the applicability of the liquid spreading analysis on the vertically-oriented tube geometry. In this manner, we expect that a simple surface parameter, u_sR_a, will

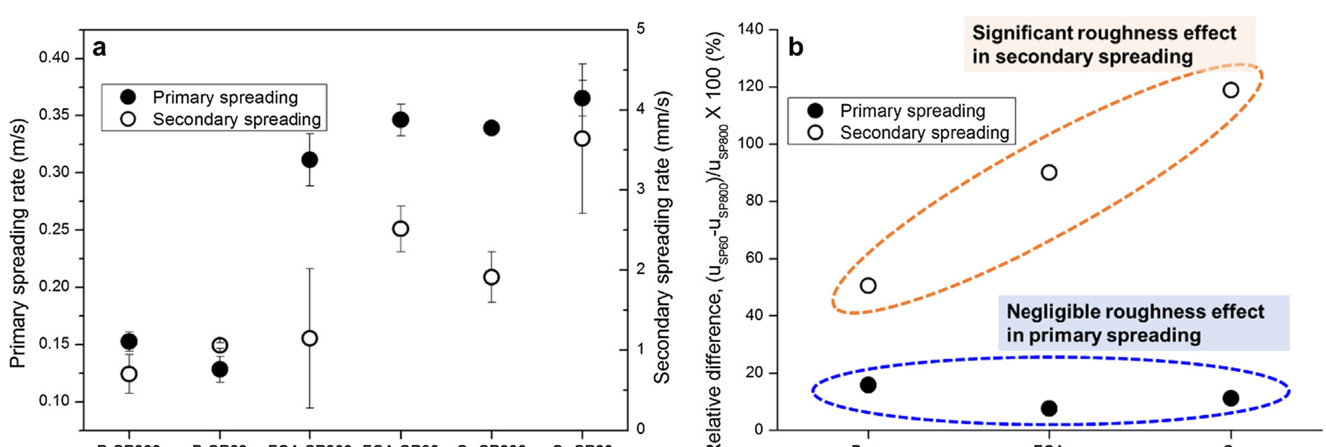

Fig. 12. (a) Variation of primary and secondary spreading rates along the explored surfaces and (b) increased roughness dependency on secondary spreading than on primary spreading.

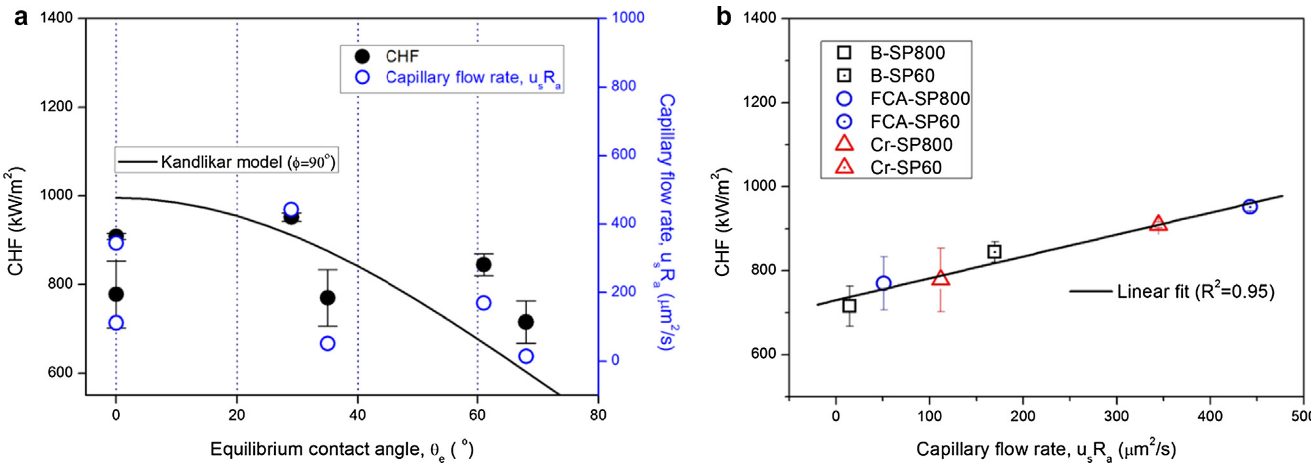

Fig. 13. (a) Improved CHF prediction with capillary flow rate compared to equilibrium contact angle and (b) linear relation between the CHF and capillary flow rate.

contribute to the analysis of the capillary wicking phenomena on the CHF enhancement of randomly-roughened tube surfaces and plate-type surfaces.

3.3. Discussion of the potential impact of the current boiling enhancement in engineering scope

This section discusses the potential functions of the different coating structures (particulate- and lumped-nanostructures) on two heater orientations (horizontal and vertical). For comparison, we referred to the pool boiling CHF of horizontal FeCrAl- [18] and Cr- [19] coated plates. The fabrication technique of the coatings and pool boiling conditions are the same for all cases.

Firstly, in Fig. 14, the NBHTC of the FeCrAl layer increased by approximately 20%, while the NBHTC of the Cr layer decreased by 10%. As discussed in Section 3.1, the NBHTC enhancement of the FeCrAl layer comes from an increase in the surface roughness. In contrast, the relatively-low roughness and superhydrophilicity of the Cr layer results in the suppression of active nucleation, which, in turn, decreases the NBHTC. This result indicates that the superhydrophilic particulate-nanostructure without microscale roughness, as shown in Fig. 2(b), does not provide good boiling efficiency in comparison to the hydrophilic lumped-nanostructure of the FeCrAl-layered surface. Thus, as a potential requirement for the surface design, certain particulate-nanostructures require removal using post-processing techniques, such as surface grinding, in order to maintain boiling efficiency.

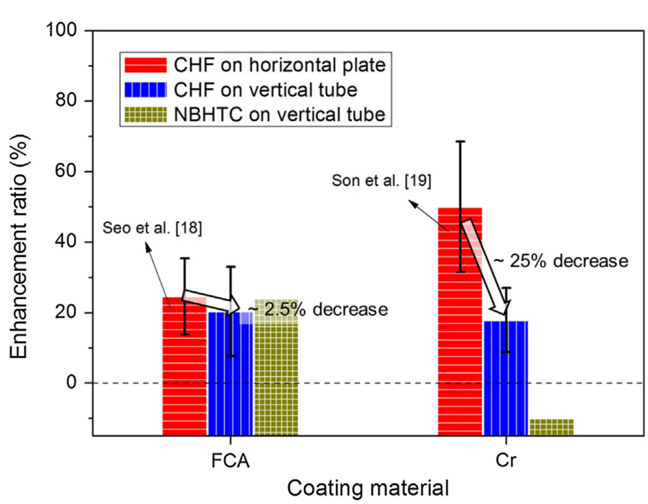

Fig. 14. Comparison of the enhancement ratios (CHF and NBHTC) for the FeCrAl and Cr coatings. This histogram only separates the coating material effect, and thus, the error bar includes the different sputtering and roughness conditions. Note that this result is generated using the magnetron DC sputtering technique.

Unlike the NBHTC, the CHF was enhanced for all of the coating materials and heater orientations under the pool boiling conditions. Thus, once wettable (superhydrophilic-to-hydrophilic) surface structures are formed at the micro/nanoscales, the consequent wetting and wicking characteristics become effective for enhancing the CHF, even at low roughness ranges (<0.3 μm in R_a). However, an important issue remains, which is the significant reduction of the CHF enhancement ratio of the superhydrophilic particulate-nanostructure when the heater orientation changes from horizontal to vertical (a CHF reduction of approximately 25% for the Cr coating is shown in Fig. 14. The CHF enhancement ratio of the hydrophilic lumped-nanostructure was generally consistent, regardless of the heater orientation. From this result, we infer that the liquid wicking effect on the particulate-nanostructures may weaken in vertical tube boiling conditions. This is potentially because the convective inertia of the bubble slug flow along the tube axial direction, which is not dominant for horizontal plate boiling, hinders the nanoscale capillary action. Although researchers [23,37,38] have consistently reported the CHF enhancement of microscale structures by changing the heater orientation from 0° to 90°, the CHF enhancement of the particulate-nanostructures seems to be diminished when compared to the noticeable enhancements of horizontal pool boiling.

Moreover, Haas et al. [39] reported that the CHF enhancement of nanoscale rough surfaces is negligible under flow boiling conditions, while structured tubes with micro-channels and a porous layer enhance the flow boiling CHF by 30%. This result indicates that the CHF enhancement generated by the nanoscale capillary action significantly decreases when the boiling condition changes from pool to flow, although the dynamics that are involved are not clear. Nevertheless, a recent RELAP5-3D simulation [40] resulted in a small amount of enhancement (~10%), which is valuable in the reducing peak cladding temperature for reactivity-initiated accidents. Thus, although significant boiling enhancement was not achieved in this study, it is essential to investigate whether the CHF can be enhanced under applicable roughness scales (<0.3 μm in R_a) and thermal–hydraulic conditions, considering its potential impact on the thermal safety of the fuel cladding system.

4. Summary and conclusions

This study investigated the surface effects of different ATF coating candidates on the pool boiling performance of vertically-oriented tubes. Using the DC magnetron sputtering technique, we fabricated FeCrAl and Cr layers (~1.5 μm thick) on different grinded tube surfaces (low roughness scale below 0.2 μm in R_a). The pool boiling performance, including the NBHTC and CHF, was compared with variations in the surface wettability and roughness. The major findings are summarized as follows:

- The hydrophilic lumped-nanostructure of the FeCrAl-layered surface resulted in the enhancement of the NBHTC and CHF. In contrast, the superhydrophilic particulate-nanostructure of the Cr-layer deteriorated the NBHTC, while enhancing the CHF. The decrease of the NBHTC on the Cr-layered surface may be attributed to the reduction of the active nucleation generated by the remarkable increase in the nucleation pressure.
- The FeCrAl and Cr coatings resulted in similar CHF enhancements of approximately 30%. The CHF trend was predictable in terms of the capillary flow rate, u_sR_a. Based on this relation, it is understood why the CHF of the FeCrAl coating (hydrophilic) is comparable with that of the Cr coating (superhydrophilic). This is because the higher roughness of the FeCrAl coating contributes to a higher amount of liquid supply, despite the lower wicking momentum than that of the Cr coating.
- As the heater orientation changes from horizontal to vertical, a significant reduction of the CHF enhancement was found for the Cr coating. The involved dynamics are not clear, but we infer that the nanoscale capillary action weakens as the convective inertia of the cooling liquid and bubble slug develops.
- In conclusion, this study reports that the FeCrAl coating (the hydrophilic lumped-nanostructure) is better than the Cr coating (the superhydrophilic particulate-nanostructure) for the boiling enhancement of the vertically-oriented tube orientation when the DC sputtering technique is used. This result is not because of the superiority of the material; it is because of the morphological difference, as other coating or post-surface-processing techniques facilitate diversified surface structures even with the same coating material.

Conflict of interest

We declare that the submitted work has no conflict of interest with any relevant research institutes, academics, or industries.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.ijheatmasstransfer.2018.09.013.

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